CMP263 - Machine Learning Paper
Japanese House Prices Predition

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Abstract.

Resumo.

1. Introduction

The Japanese housing market presents an anomaly within global real estate economics. In most developed nations, residential property functions as an appreciating asset and a means for generational wealth accumulation. In contrast, housing in Japan behaves as a depreciating consumer durable. Within 20 to 30 years of construction, the structural value of a typical Japanese home depreciates to virtually zero. This structural depreciation creates a stark institutional decoupling between land value and building value [Yoshida 2016]. While urban land, especially within the Tokyo and Osaka metropolitan areas, retains substantial value and responds to conventional market forces, the structures erected upon that land are treated as temporary assets. According to data from the Ministry of Land, Infrastructure, Transport and Tourism (MLIT), the secondary housing market accounts for only approximately 15% of total housing transactions in Japan, compared to over 80% to 90% in the US and UK [Ministry of Land, Infrastructure, Transport and Tourism 2019].

This environment traces its origins back to the post-World War II reconstruction era. Facing severe housing shortages and rapid urbanization, the Japanese government prioritized speed and quantity over long-term structural durability [Hirayama and Ronald 2006]. The resulting regulatory framework and market incentives codified a system of high-turnover construction, which was later exacerbated by the asset price bubble of the late 1980s. Japan's geographic vulnerability to seismic activity also dictates its building philosophies. The Building Standard Act of Japan has undergone continuous revisions, with the most critical shift occurring in 1981 with the introduction of the *Shin-Taishin* (New Anti-Seismic Design Code). Homes constructed prior to 1981 are fundamentally viewed as structurally unsafe and frequently fail to qualify for standard bank financing, as they're heavily penalized under modern insurance structures. Subsequent amendments following the 1995 Hanshin-Awaji (Kobe) earthquake and the 2011 Tohoku earthquake have further accelerated the technological obsolescence of older properties. Consequently, buyers often perceive purchasing a pre-owned home not just as an economic compromise, but as a direct compromise on structural safety.

Complementing these rigid safety mandates is a pervasive cultural preference for new builds, known as *Shinchiku*. A newly built home carries a significant socio-cultural premium in Japan, symbolizing clean slates, modernization, and social status. This preference is structurally reinforced by financial institutions, whereas Japanese banks routinely structure residential mortgages around the statutory useful life (*Taikyusei*) of the building.

For a standard timber-frame home, this period is legally defined as 22 years. Once a structure surpasses this threshold, securing a mortgage for its resale becomes exceedingly difficult, effectively freezing the secondary market.

It is also important to note that one of the most pressing modern challenge facing the Japanese housing market is the nation's demographic trajectory. Japan is characterized by a rapid aging society combined with a beneath replacement fertility rate, a critical example of most modern post-industrial societies. The population has been in a continuous state of decline since its peak in the late 2000s. This population contraction has triggered the abandoned house (*akiya*) phenomenon. Millions of homes across rural, suburban, and even peripheral urban areas stand completely vacant, given the tendency of inheriting families to frequently abandon the physical structures. This has created a massive inventory of valueless, deteriorating properties that distort local real estate valuations and create severe municipal management challenges.

The Japanese housing market presents an invaluable case study for urban economists and sociologists alike. It challenges the universal assumption that residential property is an inherently safe, appreciating asset class. As other post-industrial nations begin to face similar demographic plateaus and climate-induced regulatory pressures, understanding the mechanisms of Japan's "scrap-and-build" housing model becomes essential.

2. Research Problem and Objectives

This work investigates the extent to which transaction-level information can explain and predict residential real estate prices in Japan. Using data collected by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) between 2005 and 2019 and made available through Kaggle [Nishio 2019], the study frames Japanese housing valuation as a supervised regression problem. The central task is to estimate property transaction prices from observable attributes such as location, property type, land characteristics, building age, and temporal market conditions.

The problem is relevant because Japanese real estate prices are shaped by a combination of economic, demographic and geographic factors that differ from those observed in many other developed housing markets. In this context, predictive modeling is not only a technical exercise, but also a way to examine which variables carry the strongest explanatory power in a market characterized by structural depreciation, regional imbalance, and changing demand. Accordingly, this paper aims to develop, evaluate, and interpret Machine Learning regression models capable of predicting housing prices while providing insight into the main determinants of valuation in the Japanese residential market.

This study followed a reproducible empirical workflow that combined data cleaning, feature preparation, model comparison, and interpretability analysis. Several regression algorithms with distinct inductive biases, including Gradient Boosting, Random Forests, ElasticNet regression, and Support Vector Regression, were trained and evaluated under the same temporal validation strategy. Their predictive performance was compared using standard regression metrics, while feature importance and SHAP-based analyses were used to relate model behavior back to the economic and spatial determinants of Japanese housing prices.

3. Exploratory Data Analysis

The exploratory analysis began with an inspection of the structure, completeness, and temporal distribution of the MLIT transaction dataset. Each record describes a real estate transaction in Japan, with *TradePrice* defined as the target variable to be predicted. The explanatory variables combine geographic identifiers, such as prefecture, municipality, district, and nearest station, with accessibility measures, including the minimum and maximum travel time to the closest station.

The dataset also includes physical property attributes, such as area, land shape, frontage, floor plan, building year, structure, total floor area, and renovation status, as well as urban-planning information, including zoning, coverage ratio, floor-area ratio, road direction, and road classification. Temporal variables, represented by transaction year, quarter, and period, were used to examine the distribution of observations over time and to support the chronological modeling strategy adopted later in the pipeline. This initial analysis also guided the treatment of missing values, redundant variables, and features that could leak information from the target.

The correlation analysis in Figure 1 highlights the strongest linear relationships between *TradePrice* and the available numerical attributes. Property size, represented by *Area* and *TotalFloorArea*, has a clear positive association with transaction price, confirming that larger properties tend to command higher market values. Road frontage and other access-related variables are also related to price, suggesting that physical accessibility and road exposure contribute to property valuation. One less intuitive result is the weak negative correlation between *BuildingYear* and *TradePrice*, since newer buildings are usually expected to be more valuable. This pattern may reflect the fact that many older properties in the dataset are larger, renovated, or located in more established and expensive urban areas. The following plots examine these hypotheses by looking at the temporal coverage of the data, the relationship between building age and property area, and the renovation profile of older buildings.

Figure 2 summarizes the temporal structure of the dataset. The transaction-year distribution indicates how observations are spread across the market period covered by the data, while the construction-year distribution shows the age profile of the traded building stock. This is important because the project uses a chronological train-validation-test split, and because building age may capture both depreciation and location effects.

Figure 3 suggests that older buildings tend to have larger floor areas. This helps explain why the direct correlation between construction year and price can be weak, since building age is not isolated from other value drivers, especially size and location. In this dataset, some older properties may remain expensive because they offer more usable space or are concentrated in mature urban districts.

4. Methodology

4.1. Data Cleaning and Feature Preparation

Data cleaning was performed only after the chronological train-validation-test split, so that all preprocessing parameters were estimated from past observations and then applied to later periods. This design reduced the risk of information leakage from future transactions into the training process. The first step removed variables that were either not suitable

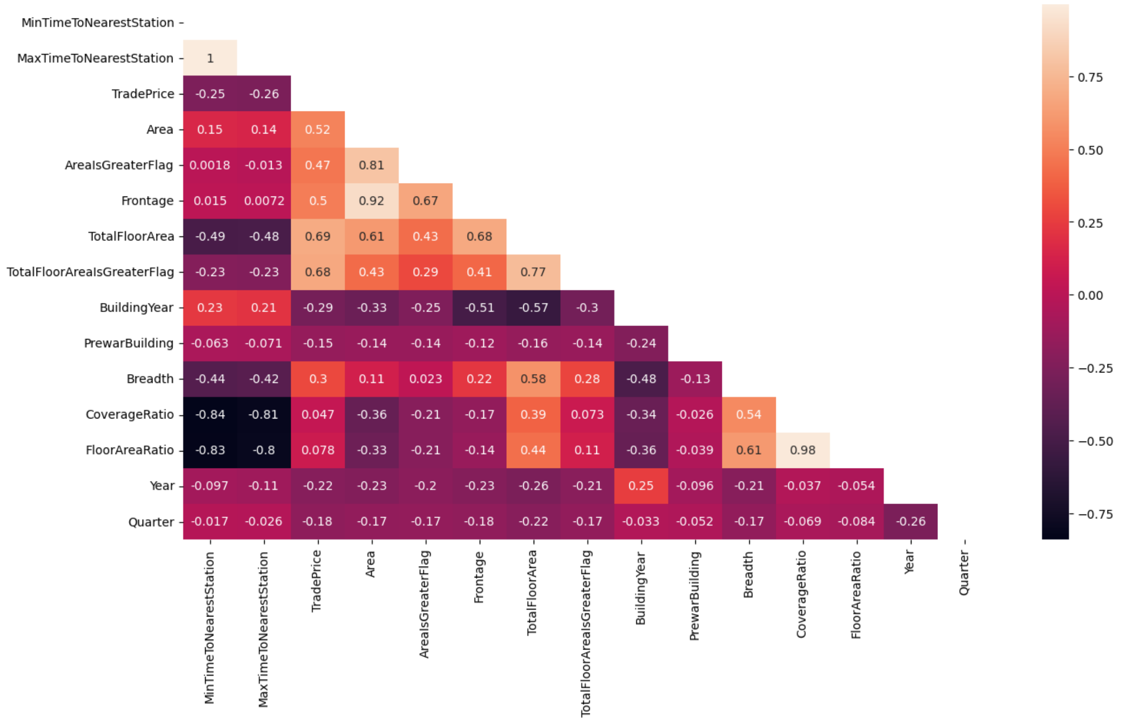

Figura 1. Correlation matrix for the numerical variables

for modeling or directly derived from the target. The textual *Remarks* field was discarded because of its unstructured nature and high missingness, same as for *Renovation*, while *UnitPrice* and *PricePerTsubo* were removed because they encode transformations of *TradePrice* and would therefore leak information about the response variable.

Missing values were then treated through a combination of domain rules and train-fitted statistical imputations. For property types in which some attributes were conceptually absent, such as land-only or rural transactions, variables including floor plan, status, structure, frontage, and nearest station were filled as not applicable or with structurally meaningful values. The original station-distance field was also reconstructed from the minimum and maximum time-to-station variables, producing a consistent numerical accessibility measure. Remaining numerical gaps were imputed with statistics learned exclusively from the training set, primarily district-level aggregates with global medians as fallbacks. After latitude and longitude were merged into the data, missing coordinates were handled with municipality-level medians, again fitted only on the training partition and then applied consistently to the validation and test sets.

Additional feature preparation was applied after the initial cleaning stage. The original station-distance representation was converted into a numerical *TimeToNearestStation* variable using the midpoint between the minimum and maximum reported travel times, while the transaction quarter was encoded with sine and cosine transformations to preserve its cyclical structure. Geographic information was enriched by merging latitude and longitude from the auxiliary locations database. The target variable was transformed with $\log(1+x)$ because housing prices are strongly right-skewed, with a small number of very expensive transactions that can dominate loss functions based on absolute or squared errors. Modeling the logarithm of price compresses these extreme values and makes the

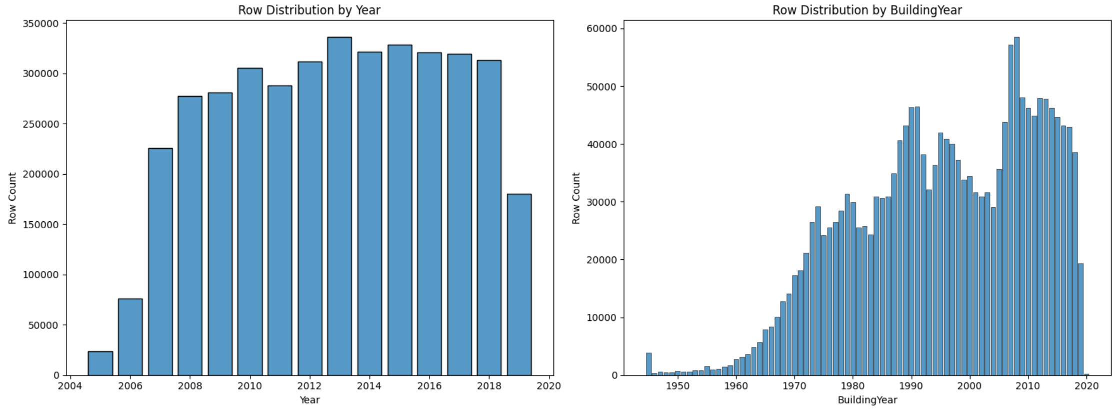

Figura 2. Temporal distributions in the dataset: transaction years and construction years of the traded properties.

learning problem closer to predicting relative price differences rather than only fitting the largest transactions. The models were therefore trained and produced predictions on the transformed scale. Before reporting evaluation metrics, both predictions and observed values were converted back to Yen using the inverse transformation. Finally, categorical variables were one-hot encoded using a layout learned on the training set, numerical features were normalized with train-fitted min-max scalers, and feature names were standardized to ensure compatibility with the different modeling libraries.

4.2. Models and Evaluation

The study compared five regression models with different assumptions and learning architectures. *XGBoost* [Chen and Guestrin 2016] was used as a gradient boosting model that builds weak decision trees sequentially, using previous errors to construct a stronger ensemble, while *LightGBM* [Ke et al. 2017] provided a more computationally efficient boosting approach based on leaf-wise tree growth. *Random Forest* represented a bagging-based ensemble of decision trees trained on bootstrapped samples, reducing variance by averaging many independent predictors. For technical reasons, the version of *Random Forest* implemented was the one from *cuML* [Raschka et al. 2020], given their faster training and inference on GPU. As a linear baseline, *ElasticNet* [Zou and Hastie 2005] combined L1 and L2 regularization to balance feature selection and coefficient shrinkage. Finally, *Support Vector Regression* [Smola and Schölkopf 2004] estimated a regression function within an error-tolerant margin, allowing nonlinear patterns to be modeled through kernel transformations.

Model performance was evaluated using complementary regression metrics computed on the original price scale after reversing the logarithmic target transformation. Mean Absolute Percentage Error (MAPE) was used to express relative prediction error, while Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) measured absolute deviations in Yen, with RMSE assigning greater weight to large errors. The coefficient of determination (R^2) was also reported to summarize the proportion of price variance explained by each model.

Reproducibility was addressed by centralizing the experimental configuration, fixing random seeds for Python and NumPy, and applying the same chronological split,

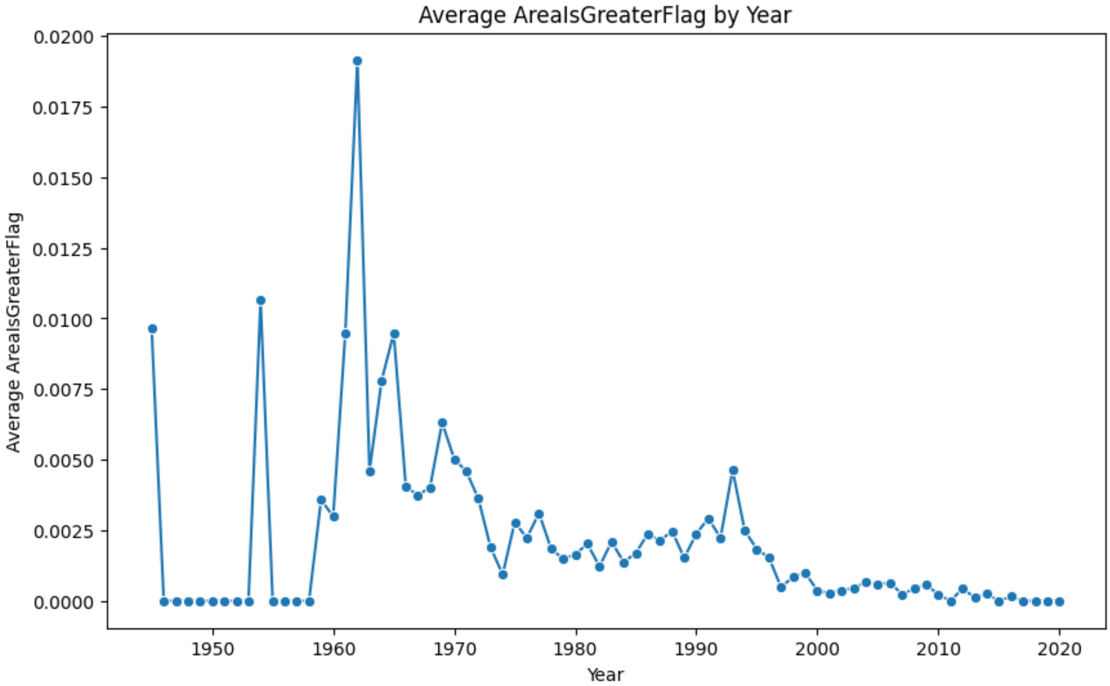

Figura 3. Relationship between building age and property area, showing that older properties often correspond to larger buildings in the observed transactions.

preprocessing rules, encoding, scaling, and model evaluation procedure across all algorithms. The pipeline also generated a run identifier and stored metadata, model input splits, and validation and test metrics, allowing each experiment to be traced back to the data partitions and feature set used during training.

Interpretability was incorporated after model selection (*post hoc*) in order to connect predictive performance with substantive housing-market explanations. For the best-performing model, global feature importance was used to identify the variables that contributed most strongly to prediction, while SHAP analysis provided a more detailed view of how individual features influenced model outputs across the test set.

5. Results

IN PROGRESS

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